Michael T. Connelly, Ph.D.

Postdoctoral Researcher, Marine Genomics, Ecology, and Evolution
National Institutes of Health (NIH), National Human Genome Research Institute (NHGRI)
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EDUCATION

2016–2021 **Ph.D.** Marine Biology and Ecology
University of Miami, Rosenstiel School of Marine, Atmospheric, and Earth Science

2012–2016 **B.Sc.** Marine Science, Biology, and Chemistry
University of Miami, Rosenstiel School of Marine, Atmospheric, and Earth Science
Magna cum laude, 2016 Marine Science Department Outstanding Student Award

RESEARCH EXPERIENCE

2023–present **Postdoctoral Fellow**, National Institutes of Health (NIH), National Human Genome Research Institute (NHGRI), Bethesda, MD
Supervisor: Dr. Andy Baxevanis
Project: "Seascape genomics and single-cell RNA-seq of the colonial marine hydrozoans *Hydractinia symbiolongicarpus* and *Podocoryna americana*"
Techniques: Genome assembly and annotation, population genomics, scRNA-seq

2021–2023 **Biodiversity Genomics Fellow**, National Museum of Natural History (NMNH), Washington, D.C., and Smithsonian Tropical Research Institute (STRI), Panama
Co-advisors: Dr. Andrea Quattrini (NMNH) and Dr. Sean Connolly (STRI)
Project: "Host-microbe coevolution and cryptic speciation in Pacific *Pocillopora* corals"
Techniques: Phylogenomic species delimitation, metagenomics, scientific SCUBA diving in Panama, museum specimen curation in Washington, D.C.

2016–2021 **Ph.D. Graduate Researcher**, RSMAES Cnidarian Immunity Laboratory
Advisor: Dr. Nikki Traylor-Knowles
Dissertation: "Interactions between the *Pocillopora* coral innate immune system and coral-associated bacteria communities"
Techniques: RNA-seq and 16S microbiome analysis, FACS and cell biology assays

2014–2016 **Assistant Researcher**, RSMAES Fish Disease Processes Laboratory
Advisor: Dr. Michael Schmale
Honors thesis: "Effects of nickel (II) chloride on zebrafish embryonic cell cultures"
Techniques: Zebrafish husbandry, cell culture, fluorescence microscopy, toxicology

2014–2016 **NOAA Hollings Scholar**, NOAA Southeast Fisheries Science Center
Advisor: Dr. Margaret Miller
Internship report: "Genet and habitat effects on outplanted *Acropora palmata* fragment growth, survivorship and bleaching response"
Techniques: Coral restoration strategic planning, photographic monitoring, quantitative image analysis, coral gamete collection in the Florida Keys National Marine Sanctuary

PUBLICATIONS

2026 (13) Chang, E.S.*, **Connelly, M.T.***, Travert, M. Barreira, S.N., Rivera, A.M., Katzer, A.M., Yu, R., Cartwright, P., Baxevanis, A.D. “A chromosome-level reference genome for the colonial marine hydrozoan *Podocoryna americana*” *G3: Genes, Genomes, Genetics* https://doi.org/10.1093/g3journal/jkag160 *co-first authors

2025 (12) **Connelly, M.T.**, Glynn, V.M., Cornejo, A., Villalobos-Cubero, T., Kleypas, J., Alvarado, J.J., Brandt, M., Logan, C.A., Palacio-Castro, A.M., Flot, J-F., Leray, M., Connolly, S.R., Quattrini, A.M. “Genome skimming illuminates hidden species diversity and Symbiodiniaceae associations in east Pacific *Pocillopora* corals” *Genome Biology and Evolution* https://doi.org/10.1093/gbe/evaf235

2024 (11) McRae, C.J., Holloway, N., Chen. G., **Connelly, M.T.**, *et al.* “Groundtruthing assessments of lab-based coral thermal tolerance with large-area imaging” *Coral Reefs* https://doi.org/10.1007/s00338-024-02582-w

2024 (10) **Connelly, M.T.**, Catapang, M.G. and Quattrini, A.M. “Unlocking the treasure trove: leveraging dry coral specimens for museum genomics” *Coral Reefs* https://doi.org/10.1007/s00338-024-02525-5

2023 (9) **Connelly, M.T.**, Snyder, G.A., Palacio-Castro, A.M., Gillette, P.R., Baker, A.C. and Traylor-Knowles, N.T. “Antibiotics reduce *Pocillopora* coral-associated bacteria diversity, decrease holobiont oxygen consumption, and activate immune gene expression” *Molecular Ecology* 32 (16), 4677-4694 https://doi.org/10.1111/mec.17049

2022 (8) **Connelly, M.T.**, McRae, C.J., Liu, P.J., Martin, C.E. and Traylor-Knowles, N.T. “Antibiotics Alter *Pocillopora* Coral-Symbiodiniaceae-Bacteria Interactions and Cause Microbial Dybiosis During Heat Stress” *Frontiers in Marine Science* https://doi.org/10.3389/fmars.2021.814124

2021 (7) Snyder, G.A., **Connelly, M.T.**, Eliachar, S., Girshoni-Yahalom, O., Browne, W.E., Palmer, C.V., Rosental, B., and Traylor-Knowles, N.T. “Functional Characterization of Hexacorallia Phagocytic Cells” *Frontiers in Immunology* https://doi.org/10.3389/fimmu.2021.662803

2021 (6) Traylor-Knowles, N.T., **Connelly, M.T.**, Young, B.D., Eaton, K., Muller, E., Paul, V.J., Ushijima, B., DeMerlis, A., Drown, M. K., Goncalves, A., Kron, N., Martin, C.E., and Rodriguez, K. “Gene Expression Response to Stony Coral Tissue Loss Disease Transmission in *M. cavernosa* and *O. faveolata* from Florida” *Frontiers in Marine Science* https://doi.org/10.3389/fmars.2021.681563

2021 (5) Nowotny, J.D., **Connelly, M.T.**, and Traylor-Knowles, N.T. “Novel Methods to Establish Whole-Body Primary Cell Cultures for the Cnidarians *Nematostella vectensis* and *Pocillopora damicornis*” *Scientific Reports* https://doi.org/10.1038/s41598-021-83549-7

2020 (4) Bonacolta, A.M., **Connelly, M.T.**, Rosales, S., del Campo, J., and Traylor-Knowles, N.T. “Microniche sampling of the bacteriome in the Starlet Sea Anemone, *Nematostella vectensis*, reveals a compartment-specific dominance of Spirochetes” *FEMS Microbiology Ecology Letters* https://doi.org/10.1093/femsle/fnab002

2020 (3) Walters, B., **Connelly, M.T.**, Young, B., and Traylor-Knowles, N.T. "The Complicated Evolutionary Diversification of the Mpeg-1/Perforin-2 Family in Cnidarians" *Frontiers in Immunology* (11):1690, https://doi.org/10.3389/fimmu.2020.01690

2020 (2) **Connelly, M.T.**, McRae, C.J., Liu, P.J., and Traylor-Knowles, N.T. "Lipopolysaccharide treatment stimulates *Pocillopora* coral genotype-specific immune responses but does not alter coral-associated bacteria communities" *Developmental and Comparative Immunology* (109):103717, https://doi.org/10.1016/j.dci.2020.103717

2017 (1) Traylor-Knowles, N.T. and **Connelly, M.T.** "What Is Currently Known About the Effects of Climate Change on the Coral Immune Response?" *Current Climate Change Reports* (3):252-260, https://doi.org/10.1007/s40641-017-0077-7

Management reports:

2015 (1) Pausch, R., Miller, M., **Connelly, M.T.**, Williams, D.E., "Effects of habitat and genet on outplanted *Acropora palmata* fragments' survival and performance" *NOAA Southeast Fisheries Science Center, in collaboration with Coral Restoration Foundation*

RESEARCH FUNDING

2025 Smithsonian Associate Director for Science and Chief Scientist (ADCS) Fiscal Year 2025 Award ($59,378, NMNH sub-award $15,644)

2022 Smithsonian Institution Barcoding Network (SIBN) Research Grant ($9,837)

2021 Smithsonian Institution Biodiversity Genomics Postdoctoral Fellowship ($118,000)

2020 University of Miami Institute for the Americas Field Research Grant ($2,000)

2018 RSMAES David Rowland Endowed Fellowship ($3,500)

2018 Marine Aquarium Societies of North America Graduate Scholarship ($5,000)

2017 RSMAES Graduate Career Development Fund for CompBio-athon ($2,500)

2016 NSF East Asia and Pacific Summer Institute Taiwan Fellow ($5,000)

2015 RSMAES Small Undergraduate Research Grant ($2,000)

2014 NOAA Ernest F. Hollings Scholarship ($8,000)

AWARDS

2020 RSMAES Marine Biology and Ecology Department Best Student Seminar ($150)

2020 UM Three-Minute Thesis Competition People's Choice Award ($50)

2018 Global Invertebrate Genomics Alliance Student Travel Award ($800)

2016 Outstanding Undergraduate RSMAES Student Award

2016 NSF Graduate Research Fellowship Honorable Mention

2015 American Chemical Society Undergraduate Award in Analytical Chemistry

2015 Barry M. Goldwater Scholarship Honorable Mention

TEACHING EXPERIENCE

2022 **Workshop Coordinator**, Coral Genetics for Conservation and Restoration Workshop
- Developed, organized, and taught a 4-day workshop on coral DNA extraction and mitochondrial barcoding techniques to Costa Rican university students and conservation professionals in San José in collaboration with Raising Coral Costa Rica.

2020 **Workshop Coordinator**, Stony Coral Tissue Loss Disease Transcriptomics Workshop
- Developed and taught a 4-day virtual workshop on coral disease RNA-seq and transcriptome analysis to RSMAES and UMiami Biology department graduate students, resulting in a collaborative, peer-reviewed publication in *Frontiers in Marine Science*.

2020 **Graduate Teaching Assistant**, Marine Ecology of the Galapagos (MSC 422)
- Supervised classroom sessions and field trips to Isabela and Floreana islands, delivered lectures on Galapagos coral ecology, ecological data collection, and introductory statistics, and mentored students through completing independent field research projects.

2018 **Graduate Teaching Assistan**t, Statistics for Environmental Management (RSM 612)
- Hosted weekly office hours, graded homework, and provided course support, including teaching introductory statistics and R programming.

2018 **Guest Lecturer**, Introduction to Comparative Immunology (MSC 465)
- Delivered invited lecture on cellular and molecular techniques used in coral immunology.

2018 **Workshop Coordinator**, RSMAES CompBio-athon
- Planned and executed a professional development workshop for 15 graduate students. Hosted invited speakers from UC Davis and Princeton University to lead bioinformatics training sessions on RNA-seq and population genomics for marine conservation.

MENTORING EXPERIENCE

2026 Ralph Cerbone, NGHRI summer research intern
2024 Jonah Petty†, NGHRI summer research intern
2022 Esme Kline, STRI/NMNH bioinformatics research intern
2022 Mary Grace Catapang, Natural History Research Experiences (NHRE) summer intern
2019 Rachel Sandquist*, RSMAES Cnidarian Immunity Laboratory
2019 James Nowotny*, RSMAES Cnidarian Immunity Laboratory
2018 Brian Walters*, RSMAES Cnidarian Immunity Laboratory
2018 Anthony Bonacolta*, RSMAES Cnidarian Immunity Laboratory
2018 Madison Emery, RSMAES Cnidarian Immunity Laboratory
2016 Megan Howson, RSMAES Cnidarian Immunity Laboratory
2016 Jessica Daly, RSMAES Cnidarian Immunity Laboratory
2016 Alanna Wasserman, RSMAES Cnidarian Immunity Laboratory

**completed RSMAES Marine Science Department Honors thesis*

†winner of Best Student Poster Award in Computational and Systems Biology at the 2024 Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS)

CONFERENCE PROCEEDINGS

Verbal Presentations:

2026 **Connelly, M.T**., Katzer, A.M., Rivera, A.M., Ryan, W., Baxevanis, A.D. “Seascape genomics of the model cnidarian *Hydractinia* reveals population differentiation and local adaptation along the U.S. Atlantic coast” Cnidofest 2026, Woods Hole, MA

2026 Zimmermann, L.A., Bejarano, S., **Connelly, M.T.**, Connolly, S.R., Marangoni, L.F.B., Keith, I., Alvarado, J.J., García, C.F., Leray, M. “Temperature Shapes Biogeographical Patterns in the *Pocillopora*-Symbiodiniaceae Symbioses across the Eastern Tropical Pacific (ETP)” International Coral Reef Symposium, Auckland, New Zealand

2026 **Connelly, M.T.**, Rivera, A.M., Travert, M., Stoilova, M.I., Cartwright, P., Baxevanis, A.D. “Cnidarian striated muscle cell evolution and development revealed through scRNAseq of the hydrozoan *Podocoryna americana*” Society of Integrative & Comparative Biology 2026 Annual Meeting, Portland, OR

2026 Stoilova, M.I., **Connelly, M.T.**, Mah, J.L., Cartwright, P. “Inferring ‘the sister to the stinger’ using phylogenetic methods at the cell-type level” Society of Integrative & Comparative Biology 2026 Annual Meeting, Portland, OR

2025 **Connelly, M.T.**, Rivera, A.M., Travert, M., Stoilova, M.I., Cartwright, P., Baxevanis, A.D. “Cnidarian striated muscle cell evolution and development revealed through scRNAseq of polyps and medusae of the hydrozoan *Podocoryna americana*” Pan-American Society for Evolutionary Developmental Biology (PASEDB) Biennial Meeting, Miami, FL

2025 Ali, V.G., Alvarado, J.J., Fernandez-García, C., **Connelly, M.T.**, Leray, M. “A Novel eDNA Assay for Monitoring *Pocillopora* spp. Corals in the Tropical Eastern Pacific” Association for Tropical Biology and Conservation 2025 Annual Meeting, Oaxaca, Mexico

2025 **Connelly, M.T.**, Rivera, A.M., Travert, M., Stoilova, M.I., Cartwright, P., Baxevanis, A.D. “Single-cell transcriptomics of polyps and medusae of the hydrozoan *Podocoryna carnea*” Society of Integrative & Comparative Biology 2025 Annual Meeting, Atlanta, GA

2024 **Connelly, M.T.**, Rivera, A.M., Travert, M., Stoilova, M.I., Cartwright, P., Baxevanis, A.D., (2024). “Single-cell transcriptomics of polyps and medusae of the hydrozoan *Podocoryna carnea*” Cnidofest 2024, Lehigh University, Bethlehem, PA

2024 **Connelly, M.T.**, Glynn, V.M., Cornejo, A., Leray, M. Connolly, S.R., Quattrini, A.M., (2024). “Genome skimming reveals species diversity and patterns of adaptation and gene flow in east Pacific *Pocillopora* corals” 3rd Joint Congress on Evolutionary Biology, Montreal, Canada

2024 **Connelly, M.T.**, Glynn, V.M., Cornejo, A., Leray, M. Connolly, S.R., Quattrini, A.M., (2024). “Genome skimming resolves east Pacific *Pocillopora* species diversity and population differentiation” Society of Integrative & Comparative Biology 2024 Annual Meeting, Seattle, WA

2021 **Connelly, M.T.**, and Traylor-Knowles, N., (2021). “Interactions between the *Pocillopora* coral innate immune system and coral-associated bacteria communities” Frontiers in Tropical Marine and Terrestrial Microbial Ecology Symposium, virtual conference, Smithsonian Tropical Research Institute

2021 **Connelly, M.T.**, McRae, C.J. Liu, P.J., Traylor-Knowles, N., (2021). “Experimental treatment of *Pocillopora* corals with LPS stimulates genotype-specific immune responses but does not alter associated bacteria communities” International Coral Reef Symposium, Bremen, Germany

2021 Traylor-Knowles, N., Snyder, G.A., **Connelly, M.T.**, Browne, W.E., Rosental, B., (2021). "From cells to genes: the surprising immune system of *Pocillopora damicornis*" International Coral Reef Symposium, Bremen, Germany

2018 **Connelly, M.T.** McRae, C.J., Liu, P.J.,Traylor-Knowles, N., (2018). "Patterns of *Pocillopora damicornis* immune gene expression in response to antibiotics treatment, heat stress, and lipopolysaccharide exposure" Global Invertebrate Genomics Alliance Conference, Curaçao

2018 Traylor-Knowles, N., Snyder, G.A., **Connelly, M.T.**, Browne, W.E., Rosental, B., (2018). "From cells to genes: the surprising immune system of *Pocillopora damicornis*" Cnidofest Conference, University of Florida Whitney Marine Laboratory, St. Augustine, FL

2018 **Connelly, M.T.** McRae, C.J., Liu, P.J., Traylor-Knowles, N., (2018). "Differential immune gene expression of *Pocillopora damicornis* corals in response to antibiotics treatment, heat stress, and lipopolysaccharide exposure" Southeastern Ecology and Evolution Conference, Miami, FL

2016 **Connelly, M.T.**, Pausch, R., Bright, A., Williams, D.E., Miller, M. (2016). "Genet and habitat effects on outplanted *Acropora palmata* fragment growth, survivorship and bleaching response" Benthic Ecology Meeting, Portland, ME

Posters:

2024 Petty, J.A., Rivera, A.M., **Connelly, M.T.**, Baxevanis, A.D., (2024) "Comparative protein structural analysis of diverse Cnidarian allorecognition systems" Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS), Pittsburgh, PA

2024 Petty, J.A., Rivera, A.M., **Connelly, M.T.**, Baxevanis, A.D., (2024) "Comparative protein structural analysis of diverse Cnidarian allorecognition systems" NIH Summer Internship Program Research Symposium, Bethesda, MD

2022 Catapang, M.G.L., **Connelly, M.T.**, Quattrini, A.M., (2022). "Assessing historical DNA from *Pocillopora* museum specimens" NHRE Poster Symposium, Washington, D.C.

2022 **Connelly, M.T.**, Quattrini, A.M., Kline, D.I. (2022). "Towards a modern integrative taxonomy of *Pocillopora* corals: a literature synthesis and meta-analysis of genetic data" Ocean Sciences Meeting, virtual conference, Association for the Sciences of Limnology and Oceanography

2020 Sandquist, R., **Connelly, M.T.**, Traylor-Knowles, N., (2020). "Investigation of tumor necrosis factor receptor-associated factors in *Pocillopora damicornis*" RSMAS Undergraduate Poster Symposium, Miami, FL

2020 Nowotny, J.D., Snyder, G.A., **Connelly, M.T.**, Rosental, B., Traylor-Knowles, N., (2020). "Working towards the establishment of long-term cnidarian cell culture and stem cell characterization through FACS" RSMAS Undergraduate Poster Symposium, Miami, FL

2020 Bonacolta, A.B., **Connelly, M.T.**, Rosales, S., Snyder, G.A., Traylor-Knowles, N., (2020). "Organismal compartmentalization of the microbiome in the starlet sea anemone, *Nematostella vectensis*" RSMAS Undergraduate Poster Symposium, Miami, FL

2018 **Connelly, M.T.** McRae, C.J., Liu, P.J., Traylor-Knowles, N., (2018). "Patterns of *Pocillopora damicornis* immune gene expression in response to antibiotics treatment, heat stress, and lipopolysaccharide exposure" Cnidofest 2018, University of Florida Whitney Marine Laboratory, St. Augustine, FL

2016 **Connelly, M.T.**, Gibbs, P.D., Schmale, M.C. (2016). "Effects of nickel (II) chloride exposure on zebrafish (*Danio rerio*) embryonic cell cultures" RSMAS Undergraduate Poster Symposium, Miami, FL

2014 **Connelly, M.T.**, Gibbs, P.D., Schmale, M.C. (2014). "Transgenic Zebrafish Cell Culture and Cell Response to External Stimuli" University of Miami Honors Summer Research Forum, Miami FL

2013 **Connelly, M.T.**, Pontes, E., Letourneau, M., Langdon, (2013). "Understanding the combined impacts of rising temperature and atmospheric CO_2 on calcification and photosynthetic rates of *Porites* and *Acropora* corals" University of Miami Research, Creativity, and Innovation Forum, Miami, FL

INVITED PRESENTATIONS

2026 University of Maryland College Park, Biological Sciences Graduate Program invited seminar, "Speciation, Symbiosis and Cellular Diversity: Genomic Perspectives from Two Cnidarians"

2025 University of Miami RSMAES Marine Biology and Ecology invited seminar, "Speciation, Symbiosis and Cellular Diversity: Genomic Perspectives from Two Cnidarian Models"

2024 Reef Futures 2024, "Genome skimming reveals species diversity and patterns of adaptation in east Pacific *Pocillopora* corals", Tropical Eastern Pacific Working Group

2024 Smithsonian NMNH All-Science Meeting, "Genome skimming reveals species diversity and patterns of adaptation in east Pacific *Pocillopora* corals"

2022 New York Society of Cosmetic Chemists, "Sources and Impacts of Pollution on Coral Reefs"

2022 McGill University Redpath Museum, "*Pocillopora* coral host-microbe interactions and coevolution"

2021 No Bones, Department of Invertebrate Zoology, NMNH, "Investigating *Pocillopora* coral host-microbe interactions, holobiont coevolution and cryptic speciation"

2021 STRI Ecological and Evolutionary Genomics Group, "Host-microbe coevolution and cryptic speciation in eastern tropical Pacific *Pocillopora* corals"

2018 Kenting National Park Headquarters, Taiwan, "Examining the innate immunity of *Pocillopora damicornis* corals from thermally variable reefs in Kenting National Park"

OUTREACH ACTIVITIES

2026 Reef Futures 2026 Solutions Expo

- Developed station "Community planning for coral reef biorepository development".

2022 Smithsonian Science Café

- Participated in science communication workshop and delivered a presentation on coral biodiversity research at the Smithsonian Institution, "Coral Microbiomes and the Key to Resilience". (https://youtu.be/TS_mhT6FOdA?t=1660)

2021 Great American Teach-In

- Led a Zoom call with 3rd-grade students in Ruskin, FL, to answer questions about Florida's coral reefs and marine ecosystems.

2020 University of Miami Three-Minute Thesis Competition

- Presented a three-minute synopsis of dissertation research findings and won award.

2020 Subtropical Experiments: “Corals Above and Below”

- Delivered a presentation on coral conservation in South Florida to a public audience at WDNA radio studio. (https://youtu.be/ksZ_jfji86M?t=1381)

2018 Frost Museum of Science IMPACT Program Mentor

- Mentored Miami-Dade County high school students Donald and Sasha for 8-week projects.

2016 President, University of Miami Aquarium Club

- Organized educational club trips to the Mote Marine Laboratory, Georgia Aquarium, Shedd Aquarium, New England Aquarium and Ripley’s Aquarium of Canada, installed a 150-gallon reef aquarium in club meeting room.

PROFESSIONAL SERVICE

2026 Coral Restoration Consortium Biorepositories Working Group Member

2024 NIH Genome Trainee Advisory Committee (GTAC) Representative

2021 Smithsonian Senate of Scientists Postdoctoral Fellow Representative

2019 Student Leadership, Evaluation and Development Committee (SLED), 2019 - 2021

Scientific Peer Review: Animal Microbiome (1), BMC Biology (1), BMC Genomics (2), Coral Reefs (2), Ecology (1), Environmental Microbiology (1), Life: The Excitement of Biology (1), PLOS ONE (4)

PROFESSIONAL DEVELOPMENT ACTIVITIES

2022 NMNH Science Café Science Communication Workshop

2021 NMNH Virtual Species Delimitation Workshop

2020 RSMAES Stony Coral Tissue Loss Disease Transcriptomics Workshop (also coordinator)

2017 RSMAES CompBio-athon (also coordinator)

2016 Mount Desert Island Biological Laboratory Environmental Genomics Workshop

PROFESSIONAL CERTIFICATIONS

2022 Smithsonian Institution Scientific Diver, 33 career SI dives in Panamá and Galápagos, Ecuador

2019 Miami-Dade College Continuing Education Department Spanish I-V course completion

2017 Department of the Interior Motorboat Operator Certification Course

2015 American Academy of Underwater Sciences authorized diver, 66 career dives with RSMAS, NOAA, and international teams in Florida, Puerto Rico, and Taiwan

SOCIETY MEMBERSHIPS

Society for Integrative and Comparative Biology (SICB)

Pan-American Society for Evolutionary Developmental Biology (PASEDB)